# SSEE and the Hubble Tension: An Algebraic Local Screening Fraction

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**Abstract**

The Hubble tension between the CMB-inferred value $H_0^{\rm CMB} = 67.4 \pm 0.5$ km/s/Mpc and the local distance-ladder measurement $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc remains one of the sharpest anomalies in modern cosmology. Within the Structural Self-Energy Expansion (SSEE) framework, the algebraic constants $\varphi$ (golden ratio) and $\pi$ determine all parameters with zero free parameters. We show that the equation of state of the SSEE dark-energy sector (Paper 7), together with the structural constants of the algebra, fixes a *local screening fraction*

$$f_{\rm screen} \;=\; \frac{s_K}{3\,\mathcal{M}} \;=\; \frac{\pi-\varphi}{(\varphi+\pi)^2} \;\approx\; 0.06725\,,$$

where $s_K = 3\,s_{\rm DE}(1+w_0) = -3w_0(1+w_0)$ is the pressure-dilution rate derived from the background equation of state, and $\mathcal{M} = \mathcal{A}/2 = (3\varphi+\pi)/4 \approx 1.9989$ is a structural constant of the algebra. The constant $\mathcal{A} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of $\varphi$ and $\pi$. The *value* of $f_{\rm screen}$ is algebraic; its physical *mechanism* is not derived in this version. Earlier versions derived it from a disformal coupling that the model no longer contains and from a separate-universe step that read $1+w_0$ as a matter density; both derivations are withdrawn (§2.2), and the multiplicative action of $f_{\rm screen}$ on a measured rate is stated as a postulate. The lens is applied in the direction set by what is measured. The only directly measured expansion rate is the local one, $H_0^{\rm SH0ES} = 73.04 \pm 1.04\,{\rm km\,s^{-1}\,Mpc^{-1}}$ (Riess et al. 2022); it enters, and the global background rate comes out,

$$H_0^{\rm glob,IR} \;=\; H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm IR}) \;=\; 73.04 \times (1 - 0.06725) \;=\; 68.13 \text{ km/s/Mpc}\,,$$

to be compared with the dimensionless algebraic number $3(\varphi+\pi)^2 = 67.96214$ (the comparison target, not an input; Paper 1): a residual of 0.166, i.e. $0.17\sigma$, with *zero free parameters.* [1] The comparison is a coincidence of value, not an

[1] **Regime, stated once for the whole paper.** This paper works in the *infrared* regime ($M \to \infty$), where $s_K = s_K^{\rm IR} = 0.403302$ and hence $f_{\rm screen} = 0.06725$. The screening fraction of the full theory is not a separate quantity but the *sum* of this term and the ultraviolet correction supplied by Paper 10, $s_K^{\rm full} = s_K^{\rm IR} + 0.013603 = 0.416905$, giving $f_{\rm screen}^{\rm full} = f_{\rm screen} + 0.002268 = 0.069522$ and the single model value $H_0^{\rm glob} = 73.04 \times (1 - 0.069522) = 67.962142$ km/s/Mpc. There is therefore *one* $H_0^{\rm glob}$, not an infrared one and an ultraviolet one: the 68.13 used throughout this paper is the same cascade evaluated with the infrared term alone, and it is superseded by the complete value once the ultraviolet completion is included. In either case the propagated uncertainty ($\pm 0.970$ in the infrared regime, $\pm 0.968$ with the full screening fraction), inherited from the SH0ES input, dominates over the residual against the pure number.

identity: $3(\varphi+\pi)^2$ is a pure number and carries no units, whereas $H_0^{\rm glob}$ inherits $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from the measurement. The agreement is therefore an output of the cascade, never its input. This adopts the single algebraic target $3(\varphi+\pi)^2$ shared by every sector of the suite (2026 reframe). The earlier register used the CMB-mapped value $H_0^{\rm MIRA} = 67.037$, giving $H_0^{\rm local} = 71.87$ ($1.12\sigma$); this is now superseded. The 2026 $\omega_m$-direct reframe adopts the pure number $3(\varphi+\pi)^2 = 67.96214$ as the algebraic target shared by all sectors — with *no* matter factor, the CMB density following as $\Omega_{m,\rm CMB} = \omega_m/h^2$ (Paper 1). The correction operates at late times ($z < 0.15$) through the scalar kinetic energy contribution to the local expansion rate, and does not modify the sound horizon $r_d$, the CMB peak positions, or the growth factor $\sigma_8$ derived in Papers 3 and 5. The UV completion of Paper 10 sharpens the residual to $4.2\times10^{-6}$ ($H_0^{\rm glob,UV} = 67.96214$), the small-scale refinement of the same cascade.

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# 1 Introduction

The Hubble tension [Verde et al., 2019, Riess et al., 2022, Kamionkowski & Riess, 2023, Di Valentino et al., 2021, Knox & Millea, 2020] is the 4–6$\sigma$ discrepancy between measurements of the present expansion rate obtained by two independent methods:

- **Early-universe inference**: CMB anisotropies (Planck 2018, Planck Collaboration 2020) combined with acoustic oscillation physics yield $H_0 = 67.4 \pm 0.5$ km/s/Mpc under the assumption of ΛCDM.
- **Local distance-ladder**: Type Ia supernovae [Riess et al., 1998, Perlmutter et al., 1999] calibrated with Cepheids in the SH0ES programme [Riess et al., 2019, 2022] give $H_0 = 73.04 \pm 1.04$ km/s/Mpc, probing predominantly $z < 0.15$.

The SSEE framework [Almeida, Paper1,P,P,P] carries the pure number

$$3(\varphi + \pi)^2 \;\approx\; 67.96214 \qquad \text{(dimensionless)}, \tag{1}$$

which, read in km/s/Mpc, lies 1.1$\sigma$ from Planck [Almeida, Paper2]. It is a comparison target, not a derived rate: a pure number cannot be the input of a dimensionful cascade, and no model has yet derived $H_0$ from first principles. The model's rate is the output of the SH0ES cascade developed below, and on its own the target does not, at first sight, resolve the local tension.

In this paper we show that the same algebraic structure that produces $H_0^{\rm alg}$ also produces a local kinematic correction $f_{\rm screen} \approx 0.06725$ from the SSEE algebra. Applied to the measured local rate $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc, this returns a global background rate $H_0^{\rm glob,IR} = 68.13$ km/s/Mpc in the IR regime, to be compared with the algebraic number $3(\varphi + \pi)^2 = 67.96214$ (0.17$\sigma$); the model value, with the complete $f_{\rm screen}$, is 67.96214 km/s/Mpc (footnote 1). The value of $f_{\rm screen}$ combines two algebraic ingredients:

1. From Paper 7 [Almeida, Paper7]: the pressure-dilution rate $s_K = -3w_0(1 + w_0) = 3\, s_{\rm DE}(1 + w_0)$, a function of the equation of state alone.
2. The structural constant $\mathcal{M} = \mathcal{A}/2 = (3\varphi + \pi)/4$, the half of $\mathcal{A}$ in the algebraic dictionary (Paper 1). Earlier versions obtained it from the disformal null geodesic of Paper 8 [Almeida, Paper8]; that coupling is not part of the action of Paper 7, and the derivation is withdrawn (§2.2).

Together these give the exact algebraic identity $f_{\rm screen} = s_K/(3\mathcal{M})$. How $f_{\rm screen}$ acts on a measured rate is not fixed by the identity: it is stated as a postulate (§3.4), and its mechanism is open (§8).

**Organisation.** Section 2 recalls the action of Paper 7 and records what was withdrawn. Section 3 derives the algebraic identity in three steps. Section 4 interprets the result physically. Section 5 verifies consistency with the rest of the SSEE series. Section 6 states the pre-committed falsifiable prediction and its current observational status. Section 7 compares the mechanism quantitatively against early dark energy, self-interacting dark radiation, and late dark-energy transitions. Section 9 summarises.

# 2 Background: the Action and the Structural Constants

## 2.1 The SSEE action (Paper 7)

The action of SSEE (Paper 7) is a $k$-essence scalar [Armendáriz-Picón et al., 1999, 2001] minimally coupled to gravity and to matter — no potential and no dark-sector coupling:

$$S = \int d^4x\,\sqrt{-g}\,\left[\frac{M_{\rm Pl}^2}{2}R + K(X) + \mathcal{L}_m\right], \qquad K(X) = c_1 X + c_2 X^2\,, \tag{2}$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$, $c_1 < 0 < c_2$, and the ratio $c_2X/c_1$ on the background is fixed by $w_0$ with no freedom (Paper 7). Light follows the null geodesics of $g_{\mu\nu}$ ($\alpha_T = \alpha_M = \alpha_B = 0$). The structural constants used below are

$$\mathcal{A} = \frac{3\varphi + \pi}{2} \approx 3.998\,, \tag{3}$$

$$\mathcal{M} = \frac{\mathcal{A}}{2} = \frac{3\varphi + \pi}{4} \approx 1.9989\,. \tag{4}$$

Earlier versions wrote $K(X,\phi) = X/\mathrm{KAL}_0 + X^2/M^4 - V_0\,e^{-\lambda\phi/M_{\rm Pl}}$ with a conformal coupling $\beta_c \equiv \mathcal{A}$ to dark matter. Paper 7 withdrew the potential and the coupling ($\beta_c = -\mathcal{A}$ was the artefact of a normalisation error); no result of this paper depends on either.

The dark-energy equation-of-state parameters derived in Paper 1 are

$$w_0 = -\frac{\mathcal{A}}{\varphi + \pi} \approx -0.840\,, \tag{5}$$

$$s_{\rm DE} = -w_0 = \frac{\mathcal{A}}{\varphi + \pi} = 0.839950\,. \tag{6}$$

The quantity entering the screening fraction is the fractional dilution rate of the dark-energy *pressure* per e-fold,

$$s_K \;\equiv\; -\frac{1}{\rho_\phi}\frac{dp_\phi}{dN} \;=\; -3w_0(1 + w_0)\,, \qquad N = \ln a, \tag{7}$$

which carries only the equation of state: no factor of $H$, no density, and no kinetic function. Section 3 gives its exact algebraic form.

**A label corrected.** Earlier versions of this paper called this quantity $\alpha_K$ and defined it as the Bellini–Sawicki kineticity $2XK_X/(M_{\rm Pl}^2H^2)$ [Bellini & Sawicki, 2014, Gubitosi et al., 2013, Gleyzes et al., 2013]. That identification was wrong, and it matters here rather than being a matter of notation. The kineticity of the SSEE action is $\alpha_K = 3(1-\Omega_m)(5-3w_0) = 15.591336$ (Paper 7, verified against HI_CLASS); it depends on $K(X)$ and it evolves. The constant 0.403302 is $s_K$, and the distinction is what makes $f_{\rm screen}$ well defined: a screening fraction built from a kineticity would inherit that kineticity's dependence on the expansion history, while one built from $s_K$ cannot, because $s_K$ contains no $H$ (§B).

### 2.2 What was withdrawn: the disformal coupling and the separate-universe step

Earlier versions of this paper obtained $\mathcal{M}$ from Paper 8, where dark matter was coupled to the disformal metric $\tilde{g}_{\mu\nu} = g_{\mu\nu} + (2/M^4)\,\partial_\mu\phi\,\partial_\nu\phi$ [Bekenstein, 1993, Zumalacárregui & García-Bellido, 2014]; the disformal null geodesic amplified the deflection by $\beta_c$ and the Einstein radius by $\approx \mathcal{M}$, and $\mathcal{M}$ was read as the optical coupling of that sector. They also derived the multiplicative form of the correction twice: from a compression of the luminosity distance along disformal geodesics, and from a separate-universe argument [Wands et al., 2000, Brax & Valageas, 2014]. All three are withdrawn, for two reasons.

(i) The action of Paper 7 is minimally coupled. It contains neither a disformal nor a conformal coupling, and photons travel on the null geodesics of $g_{\mu\nu}$. The lensing prediction of Paper 8 is its limit with $\mu - 1 = 0$; the disformal limit is kept there only as a historical contrast. $\mathcal{M}$ is therefore not an optical coupling of anything, and no path compression exists to derive.

(ii) The separate-universe step cancelled a factor $\Omega_m/(1 + w_0)$ through the identity "$1 + w_0 = \Omega_m$". But $1 + w_0 = s_m = 0.160050$ is a number of the equation of state, not a matter density; the one matter density of the model is $\Omega_m = 0.308881$. The identity does not hold and the step fails with it. It also took a Local-Group overdensity $\delta_{\rm local} \approx 2$ as an input.

What survives is the value: $\mathcal{M} = \mathcal{A}/2$ as an algebraic constant and $f_{\rm screen} = s_K/(3\mathcal{M})$ as an identity (§3). What does not survive is any derivation of how $f_{\rm screen}$ acts on a measured rate. That action, $H_0^{\rm local} = H_0^{\rm glob}/(1 - f_{\rm screen})$, is stated in §3.4 as a postulate; two tests of its mechanism against present data are reported in §8.

## 3 Algebraic Derivation: $f_{\rm screen} = s_K/(3\mathcal{M})$

We derive the value of the screening fraction in three steps using only algebraic relations. The word "screening" is kept for continuity with earlier versions and with the literature on scalar screening [Vainshtein, 1972, Babichev & Deffayet, 2013]; here it names a fraction, not a mechanism (§2.2).

### 3.1 Step 1: Exact expression for $s_K$

With $p_\phi = w_0\rho_\phi$ and $w_0$ constant, the continuity equation $\rho'_\phi = -3(1 + w_0)\rho_\phi$ gives

$$\frac{1}{\rho_\phi}\frac{dp_\phi}{dN} = w_0\,\frac{\rho'_\phi}{\rho_\phi} = -3w_0(1 + w_0)\cdot(-1)\,, \qquad \text{hence} \qquad s_K = -3w_0(1 + w_0)\,. \tag{8}$$

Writing $s_{\rm DE} \equiv |w_0| = T_r/M_v$ for the saturation and $s_m \equiv 1 + w_0$ for its complement ($s_{\rm DE} + s_m = 1$ exactly), this is $s_K = 3\,s_{\rm DE}\,s_m$. Nothing but the equation of state enters.

Using the SSEE algebraic values (5)–(6),

$$s_{\rm DE}\,(1 + w_0) = \frac{\mathcal{A}}{\varphi + \pi}\cdot\frac{\varphi + \pi - \mathcal{A}}{\varphi + \pi} = \frac{\mathcal{A}(\pi - \varphi)/2}{(\varphi + \pi)^2}\,, \tag{9}$$

where we used $(\varphi + \pi) - \mathcal{A} = (\pi - \varphi)/2$ (exact identity from the definitions (3) and $\varphi + \pi = \Omega$):

$$\Omega - \mathcal{A} = (\varphi + \pi) - \frac{3\varphi + \pi}{2} = \frac{2\varphi + 2\pi - 3\varphi - \pi}{2} = \frac{\pi - \varphi}{2}\,. \tag{10}$$

Therefore,

$$\boxed{s_K = \frac{3\mathcal{A}(\pi - \varphi)}{2(\varphi + \pi)^2} = 0.403302458888\,.} \tag{11}$$

The closed form and $-3w_0(1 + w_0)$ agree to $1.1 \times 10^{-16}$, machine zero.

### 3.2 Step 2: Exact expression for $3\mathcal{M}$

$\mathcal{M}$ is the half of $\mathcal{A}$, equation (4):

$$\mathcal{M} = \frac{\mathcal{A}}{2} = \frac{3\varphi + \pi}{4}\,. \tag{12}$$

With the triple $T_r = 3(\varphi + \beta) = 3\mathcal{A}$ of the dictionary,

$$3\mathcal{M} = \frac{3\mathcal{A}}{2} = \frac{T_r}{2}\,. \tag{13}$$

Earlier versions derived Eq. (12) from the disformal geodesic of Paper 8; that derivation is withdrawn (§2.2) and the value stands as an algebraic definition.

### 3.3 Step 3: The cancellation and the identity

Dividing equation (11) by equation (13):

$$\begin{aligned} \frac{s_K}{3\mathcal{M}} &= \frac{3\mathcal{A}(\pi - \varphi)}{2(\varphi + \pi)^2} \cdot \frac{2}{3\mathcal{A}} \\ &= \frac{\pi - \varphi}{(\varphi + \pi)^2}\,. \end{aligned} \tag{14}$$

The factor $\mathcal{A}$ (equivalently $2\mathcal{M}$) cancels exactly. The result is a pure algebraic function of the two fundamental constants $\varphi$ and $\pi$:

$$\boxed{f_{\text{screen}} \;\equiv\; \frac{s_K}{3\mathcal{M}} \;=\; \frac{\pi - \varphi}{(\varphi + \pi)^2} \;\approx\; 0.06725\,.} \tag{15}$$

By Eq. (10) the same number reads $f_{\text{screen}} = 2(\Omega - \mathcal{A})/\Omega^2 = (K_v - 2\mathcal{A})/\Omega^2$, with $K_v = \varphi + \pi + \Omega = 2\Omega$: twice the gap between $\Omega$ and $\mathcal{A}$, in units of $\Omega^2$. This is the central result of the paper. No parameters are adjusted: $s_K$ is fixed by the equation of state; $\mathcal{M}$ is a structural constant; their ratio is fixed by $\varphi$ and $\pi$. The identity fixes the *value* of $f_{\text{screen}}$; it does not by itself say how $f_{\text{screen}}$ acts on a measured rate (§3.4).

## 3.4 The form of the correction: a postulate

The identity fixes the value of $f_{\rm screen}$. Which form the correction takes on a measured rate — multiplicative or additive — is not fixed by it. Earlier versions derived the multiplicative form from a separate-universe argument and from a disformal compression of the luminosity distance; both are withdrawn (§2.2). We therefore adopt the multiplicative form as the *screening postulate*:

$$H_0^{\rm local} \;=\; \frac{H_0^{\rm glob}}{1 - f_{\rm screen}}\,. \tag{16}$$

It is a statement about observables, not a derivation: Appendix B spells out what it implies for the luminosity distance, and Section 6 lists the observations that can falsify it. The additive variant $H_0^{\rm glob}(1 + f_{\rm screen})$ differs by $O(f_{\rm screen}^2) \approx 0.31$ km/s/Mpc, below the present SH0ES uncertainty, so the choice moves no conclusion of this paper.

## 3.5 Numerical verification

Table 1 lists all intermediate numerical values.

Table 1: Numerical verification of the algebraic identity (15).

| Quantity | Algebraic expression | Numerical value | Source |
|---|---|---|---|
| $\varphi$ | $(1+\sqrt{5})/2$ | 1.6180339887 | Definition |
| $\pi$ | $\pi$ | 3.1415926536 | Definition |
| $\Omega = \varphi + \pi$ | — | 4.7596266423 | — |
| $\mathcal{A}$ | $(3\varphi + \pi)/2$ | 3.9978473099 | Eq. (3) |
| $\mathcal{M}$ | $\mathcal{A}/2$ | 1.9989236550 | Eq. (12) |
| $w_0$ | $-\mathcal{A}/\Omega$ | −0.83994977 | Eq. (5) |
| $s_{\rm DE}$ | $\mathcal{A}/\Omega$ | 0.83994977 | Eq. (6) |
| $1 + w_0$ | $(\pi - \varphi)/(2\Omega)$ | 0.16005023 | Eq. (10) |
| $s_K$ | $3\mathcal{A}(\pi - \varphi)/(2\Omega^2)$ | 0.403302 | Eq. (11) |
| $3\mathcal{M}$ | $3\mathcal{A}/2$ | 5.9968 | Eq. (13) |
| $f_{\rm screen} = s_K/(3\mathcal{M})$ | $(\pi - \varphi)/\Omega^2$ | 0.067253 | Eq. (15) |
| $H_0^{\rm SH0ES}$ (*input*, measured) | — | $73.04 \pm 1.04$ km/s/Mpc | Riess et al. [2022] |
| $H_0^{\rm glob,IR}$ (Paper-9 IR regime, mult.) | $H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm IR})$ | 68.13 km/s/Mpc | Eq. (18) |
| $H_0^{\rm glob}$ (**model value**, mult.) | $H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm full})$ | **67.96214** km/s/Mpc | Paper 10, Eq. (sKfullform) |
| $H_0^{\rm glob,IR}$ (additive variant) | $H_0^{\rm SH0ES}/(1 + f_{\rm screen}^{\rm IR})$ | 68.44 km/s/Mpc | — |
| $3\Omega^2$ (pure number, target) | $3(\varphi + \pi)^2$ | 67.96214 | Paper 1 (target, not input) |
| $H_0^{\rm CCHP}$ (TRGB, adopted) | — | $70.39 \pm 1.94$ km/s/Mpc | Freedman et al. [2025] |
| Residual from SH0ES (IR regime) | — | $0.17\sigma$ | — |
| Residual from CCHP (IR regime) | — | $-1.28\sigma$ | — |
| *superseded:* $H_0^{\rm MIRA}$=67.037 → 71.87 | MIRA mapping | $1.12\sigma$ | old register |

**A figure withdrawn.** Earlier versions plotted $f_{\rm screen}(z)$, obtained by scaling $s_K$ with a dark-energy fraction built from $\Omega_m = 1 - s_{\rm DE} = 0.160050$. That number is $1 + w_0$, not a density (§2.2), and the curve contradicted Step 4 of Appendix B: $f_{\rm screen}$ contains only $w_0$ and is constant at all $z$. The figure is withdrawn.

# 4 Physical Interpretation

## 4.1 Two-epoch structure of $H_0$ measurements

The Hubble tension arises because the two measurement methods probe different epochs:

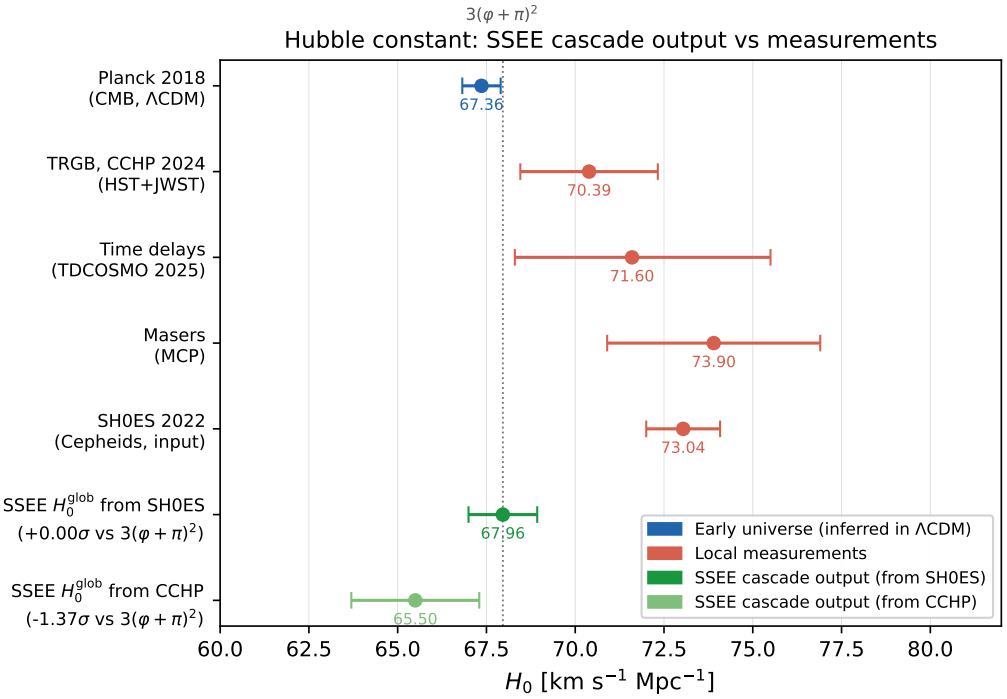



Figure 1: Hubble constant measurements: the early-universe inference (blue, within ΛCDM) and local measurements (red), each read from its publication, against the output of the SSEE cascade with the complete screening fraction (green). Run on SH0ES the cascade returns $H_0^{\rm glob} = 67.96214$ km/s/Mpc; run on the TRGB value adopted by CCHP [Freedman et al., 2025] it returns 65.50 ($-1.37\sigma$ from $3(\varphi+\pi)^2$), the control from the other side. Error bars show $\pm1\sigma$.

- The CMB infers $H_0$ via the angular scale $\theta_* = r_d/D_A(z_*)$ at $z_* \approx 1100$, where the SSEE dark energy density is $\Omega_{\rm DE}(z_*) \sim 10^{-9}$ and the scalar kinetic energy is negligible. The inferred value is $H_0^{\rm glob}$, which coincides numerically with the pure number $3(\varphi+\pi)^2$.
- SH0ES measures $H_0$ via the distance ladder at $z < 0.15$, where $s_{\rm DE} \approx 0.840$ and the scalar field is kinetically active.

The kinetic part of the scalar energy density is

$$\rho_{\rm kin} \equiv XK_X = \frac{\rho_\phi + p_\phi}{2} = \frac{\rho_\phi(1+w_0)}{2}\,, \tag{17}$$

which is active at late times and negligible at $z_*$, where the dark energy is a negligible fraction of the total. Earlier versions continued this equation with $\rho_\phi = 3M_{\rm Pl}^2H^2 s_{\rm DE}$, writing the saturation $s_{\rm DE} = |w_0|$ in the place of a density fraction, and so identified $f_{\rm screen}$ with a share of the energy budget; that step is withdrawn. Eq. (17) only says that the scalar is kinetically active where the ladder measures and not where the CMB is set; it is not the mechanism of $f_{\rm screen}$.

## 4.2 Local Hubble correction

The screening fraction $f_{\rm screen} = s_K/(3\mathcal{M})$ is a number; how it enters a measured rate is the postulate of §3.4. In its multiplicative form, and for comparison in the additive one,

the cascade gives:

**Which end is the input.** The lens is a pure number: $f_{\rm screen}$ is built from $s_K$ and $\mathcal{M}$, both fixed by the equation of state and by $\varphi, \pi$ alone, so it contains no factor of $H$ and no density. It does not ask which $H$ enters — it converts one measured rate into the other. The direction is therefore set by what is actually measured, and only one expansion rate is: the local one. The SH0ES distance ladder [Riess et al., 2022] yields $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc from Cepheid-calibrated supernovae. The Planck value is not an alternative input of the same kind: it is inferred rather than measured, and inferred within ΛCDM, whose free parameters are adjusted to the data — so the $H_0$ it reports is where those free parameters land, not an independent determination. We therefore take SH0ES as the input throughout:

$$H_0^{\rm glob} \;=\; H_0^{\rm SH0ES}\,(1-f_{\rm screen}) \;=\; 73.04\times\left[1-\frac{\pi-\varphi}{(\varphi+\pi)^2}\right] \;=\; 68.13\ \text{km/s/Mpc} \quad [\text{multiplicative}]\,, \tag{18}$$

$$H_0^{\rm glob} \;=\; \frac{H_0^{\rm SH0ES}}{1+f_{\rm screen}} \;=\; 68.44\ \text{km/s/Mpc} \quad [\text{additive}]\,. \tag{19}$$

The difference between the two forms is of order $f_{\rm screen}^2 \approx 0.31$ km/s/Mpc. The output is then compared with the dimensionless algebraic number $3(\varphi+\pi)^2 = 67.96214$ (the comparison target, Paper 1). The residual is 0.166 in the multiplicative form and 0.475 in the additive one, i.e. in units of the propagated SH0ES uncertainty $1.04 \times (1 - f_{\rm screen}) = 0.970$:

$$\Delta_{\rm mult} = 0.17\sigma\,, \qquad \Delta_{\rm add} = 0.49\sigma\,. \tag{20}$$

These are the same two numbers reported in earlier versions of this paper, which ran the cascade in the opposite direction; the tensions are invariant under the inversion because the lens is multiplicative. What changes is the epistemic status of the comparison. A pure number cannot serve as the input of a dimensionful cascade, so $3(\varphi+\pi)^2$ is not an anchor that the model assumes: it is a target the model either hits or misses, and the hit is the result.

**Superseded MIRA-mapped value.** The earlier register applied the same mechanism to the CMB-mapped value $H_0^{\rm MIRA} = 67.037$, giving $H_0^{\rm local} = 71.87$ ($1.12\sigma$). Under the 2026 $\omega_m$-direct reframe the suite adopts the pure number $3(\varphi+\pi)^2 = 67.96214$ as the algebraic target — with no matter factor ($\Omega_{m,\rm CMB} = \omega_m/h^2$; Paper 1) — so the cascade output is compared against 67.96214 and the MIRA-mapped value is retired as canonical.

**Control from the other side: TRGB/JWST.** The Chicago–Carnegie programme adopts $H_0 = 70.39 \pm 1.94$ km/s/Mpc from the tip of the red-giant branch [Freedman et al., 2025] (statistical, systematic and supernova terms in quadrature). Run on this datum the IR lens returns $H_0^{\rm glob,IR} = 65.66$ km/s/Mpc, a residual of $-2.31$ against $3(\varphi+\pi)^2$, i.e. $-1.28\sigma$ in units of the propagated uncertainty. Earlier versions used $69.96 \pm 1.54$, the estimate of the first arXiv version of that paper (August 2024); the revised version (March 2025) adopts the value used here, and the residual quoted then, $1.88\sigma$, is superseded with it. The SH0ES and CCHP inputs differ by about their combined uncertainty; independent local probes — strong-lensing time delays [Wong et al., 2020] and water megamasers [Pesce et al., 2020] — span $H_0 \approx 70$–$74$ km/s/Mpc, an observational spread that is independent of the SSEE framework. We report both comparisons in Table 1.

We adopt the **multiplicative form** (18) because it is the screening postulate (16), not because it is derived. Read on the observable, it says that the slope of the Hubble diagram at $z \lesssim 0.15$, which the ladder measures, is the global rate boosted by $(1 - f_{\rm screen})^{-1}$; Appendix B shows that this is the same as a uniform compression $d_L^{\rm obs} = (1 - f_{\rm screen})\, d_L^{\rm glob}$ at $O(z)$. The additive form (19) is its linearisation $(1 - f_{\rm screen})^{-1} \approx 1 + f_{\rm screen}$, with a second-order difference $O(f_{\rm screen}^2) \approx 0.31\,\mathrm{km/s/Mpc}$. Since $f_{\rm screen}$ is exactly constant (Appendix, Step 4) no average over redshift is needed.

*Note on the role of* $\mathcal{M}$. $\mathcal{M} = \mathcal{A}/2$ enters $f_{\rm screen}$ only through the denominator $3\mathcal{M} = T_r/2$, where $\mathcal{A}$ cancels against the $\mathcal{A}$ in $s_K$. Earlier versions read it as the optical coupling of a disformal sector and connected $f_{\rm screen}$ to a lensing amplification of Paper 8; with the disformal coupling withdrawn, that connection is withdrawn too, and $\mathcal{M}$ plays no role here beyond its value.

### 4.3 Why the CMB is unaffected

The correction $f_{\rm screen}$ is a late-time effect generated by scalar kinetic energy that is negligible at $z \gg 1$. Specifically:

1. **Sound horizon** $r_d$: not modified. The correction enters through $w_0$ and $s_{\rm DE}$ at $z \sim 0$, long after decoupling. Papers 2 and 3 established $r_d = 147.2\,\mathrm{Mpc}$, consistent with Planck, and this result is unchanged.

2. **CMB peak positions**: not modified. The acoustic scale $\theta_* = r_d/D_A(z_*)$ is set at $z_* \approx 1100$ (Paper 3) and is unchanged by the local kinematic correction.

3. **Growth factor** $\sigma_8$: not modified. The IS viscosity (Paper 5) operates through the growth equation, not through $H_0$, so the growth results of Paper 6 (the $S_8$ prediction against KiDS-Legacy) are unaffected.

**Note on the EDE comparison.** Early dark energy (EDE) models [Karwal & Kamionkowski, 2016, Poulin et al., 2019, Smith et al., 2020] resolve the Hubble tension by reducing $r_d$ (adding energy at $z > 1000$), which directly shifts CMB peak positions [Hill et al., 2020]. A naive application of $f_{\rm screen}$ as an EDE-like modification to $r_d$ would give $r_{d,\rm new} = r_d/(1 + f_{\rm screen}) \approx 137.3\,\mathrm{Mpc}$, displacing the first peak by $+7\%$ ($\approx 24\sigma$) — which is excluded by Paper 3. The present mechanism does *not* modify $r_d$; it operates entirely at $z < 0.15$ through the scalar kinetic energy, a physically distinct sector.

## 5 Consistency with the SSEE Series

Table 2 confirms that the correction $f_{\rm screen}$ does not retroactively modify any result in Papers 1–8. Its one input from the series is $s_K$ (Paper 7); $\mathcal{M}$ is a structural constant of the dictionary. Paper 9 derives the value of their ratio and postulates its action on the distance ladder.

## 6 Pre-committed Falsifiable Prediction

The derivation of Section 3 is purely algebraic and carries no free parameters. We therefore register the following pre-committed prediction:

Table 2: Cross-consistency of the Paper 9 result with the SSEE series.

| Paper | Key result | Affected by $f_{\rm screen}$? |
|---|---|---|
| 1 (Framework) | $3(\varphi+\pi)^2$ | No — a pure number, the target the cascade output is compared against (not an input of Postulate D); the superseded MIRA-mapped 67.037 is retired |
| 2 (MCMC DESI) | MCMC posterior $H_0 = 67.82 \pm 0.41$ (prior $H_0^{\rm glob}$) | No — MCMC measures the global background, not local |
| 3 (CMB) | $r_d = 147.2$ Mpc, peaks at $\ell = 220, 540, 810$ | No — $f_{\rm screen}$ does not shift $r_d$ or $\theta_*$ |
| 4 (Sovereignties) | $H_0^{\rm glob}$ as first sovereignty | No — $H_0^{\rm local}$ is a new sector |
| 5 (IS viscosity) | $\sigma_8 = 0.8136$, $G = D_1^{\rm SSEE}/D_1^{\Lambda{\rm CDM}} = 1.0032$ | No |
| 6 (growth) | $S_8 = 0.8273$ predicted vs. KiDS-Legacy $0.8265 \pm 0.0176$ ($0.04\sigma$; *no* free cosmological parameter) | No |
| 6 (growth, earlier) | $S_8 = 0.7559 \pm 0.0189$ vs. raw KiDS-1000 | No |
| 7 (EFT) | $s_K = 0.403302$, $c_s^2 = 0.021284$ | Provides $s_K$ (input to Paper 9) |
| 8 (Strong gravity) | Lensing with $\mu - 1 = 0$; the disformal limit is historical | No — $\mathcal{M}$ is no longer taken from Paper 8 (§2.2) |

**SSEE predicts:**

$$H_0^{\rm glob} = H_0^{\rm SH0ES}\left[1 - \frac{\pi-\varphi}{(\varphi+\pi)^2}\right] = 73.04 \times (1 - 0.06725) = 68.13 \text{ km/s/Mpc}\,, \tag{21}$$

where the output is compared against the pure number $3(\varphi+\pi)^2 = 67.96214$ (the comparison target, Paper 1), the same value the CMB sector recovers—the CMB matter density is the $\omega_m$-direct forward identity $\Omega_{m,\rm CMB} = \omega_m/h^2$, with *no* matter factor (OP-8 dissolved). *Current status*: $0.17\sigma$ from SH0ES [Riess et al., 2022].

**Falsification criteria:**

1. If future SH0ES or SHOES-equivalent measurements (post-2027) converge to $H_0^{\rm local} < 71.5$ or $> 74.2$ km/s/Mpc, the prediction is excluded at $\geq 1\sigma$.

2. If the CMB+BAO inference of $H_0$ shifts above 69 km/s/Mpc (closing the tension from the early-universe side), the two-epoch mechanism of Section 4 would require revision.

3. If the SSEE EFT is found to require $s_K \neq 3s_{\rm DE}(1+w_0)$ (e.g., from a non-standard UV completion), the identity (15) would be modified proportionally.

4. **Time-delay test with SN Requiem (registered 2026-09-19, before the data).** A lensed-supernova time delay measures the global expansion without the Cepheid ladder, so it must read $H_0^{\rm glob}$, not the local value. Because the SSEE background has $(w_0, w_a) \neq (-1, 0)$, a flat-ΛCDM analysis of the same delays returns a value higher by the ratio of time-delay distances, $\simeq 1.6\%$ for MACS J0138.0−2155, i.e. $H_0 \simeq 69.1\,\mathrm{km/s/Mpc}$. The lens models of Suyu et al. [2026] place the reappearance of the fourth image of SN Requiem in April–December 2026 for $H_0 = 73\,\mathrm{km/s/Mpc}$ and in March–November 2027 for $H_0 = 67\,\mathrm{km/s/Mpc}$; interpolating in $1/H_0$, SSEE centres it near March 2027 and the local value near August 2026 (±4 months each). A detection before 2027 favours the local value; none before 2027 favours $H_0^{\rm glob}$. The subsequent 2–3% measurement decides: a flat-ΛCDM value near 73 km/s/Mpc excludes the screening mechanism. Current lensed-supernova values [Pierel et al., 2025, Bazzanini et al., 2026] are consistent with both.

5. **Rival reading of the screening (registered 2026-10-03).** Criterion 4 assumes that the screening acts only on the Cepheid ladder, so that a lens reads the global rate. In the rival reading the screening is a property of every local measurement of the expansion rate, lenses included: the lens then reads the local value, and a flat-ΛCDM analysis of the MACS J0138−2155 delays returns $H_0 \simeq 74.2\,\mathrm{km/s/Mpc}$ instead of $\simeq 69.1\,\mathrm{km/s/Mpc}$. The fourth image of SN Requiem then reappears inside the window the lens models assign to $H_0 \approx 73$, in 2026 [Suyu et al., 2026, Pierel et al., 2026]. The same datum decides between the two readings: a reappearance in 2026 falsifies criterion 4 and leaves this reading; a reappearance in 2027 falsifies this reading. We record that it is registered late: most of its 2026 window had already elapsed without a public report of the image, so the absence so far already weighs against it, pending the results of the monitoring campaign. A comparison of the two readings with present ladder-free data (masers, lenses, sirens) lacks the power to decide.

# 7 Quantitative Comparison with Competing Resolutions

The Hubble tension has motivated a large family of proposed extensions to ΛCDM. Schöneberg et al. [2022] graded $\sim 20$ such models on a uniform set of datasets (the "$H_0$ Olympics"), establishing that no single proposal fully closes the tension without side-effects. To position the SSEE mechanism within this landscape, we compare it quantitatively against the three best-studied classes of resolution.

## 7.1 The three competing classes

**(a) Early dark energy (EDE).** EDE [Karwal & Kamionkowski, 2016, Poulin et al., 2019, Smith et al., 2020] introduces a scalar field that contributes a fraction $f_{\rm EDE} \sim 0.1$ of the energy budget near matter–radiation equality ($z_c \sim 3000$–$5000$) and then dilutes faster than radiation. By injecting energy before recombination it *reduces the sound horizon* $r_d$ by a few percent, which raises the CMB-inferred $H_0$ to $\approx 71$–$72.5\,\mathrm{km/s/Mpc}$. The cost is threefold: (i) three new parameters ($f_{\rm EDE}$, $z_c$, the initial field displacement $\theta_i$); (ii) a compensating increase in the scalar spectral index $n_s$ and the cold dark-matter density

$\omega_{\rm cdm}$, which *worsens* the $S_8$ tension [Hill et al., 2020]; and (iii) a residual $\sim 1.6$–$2\sigma$ offset from SH0ES when no $H_0$ prior is imposed.

**(b) Self-interacting dark radiation (SIDR).** Dark-radiation models add relativistic degrees of freedom ($\Delta N_{\rm eff}$), optionally self-interacting to evade free-streaming constraints. Self-interacting neutrinos [Kreisch et al., 2020] and the Wess–Zumino dark-radiation (WZDR) model [Aloni et al., 2022] are the most successful realisations, reaching $H_0 \approx 70$–$71.8\,$km/s/Mpc with one to three new parameters. These models also act *before* recombination — they modify $r_d$ and the CMB damping tail through $\Delta N_{\rm eff}$ — and leave a residual $\sim 2$–$3\sigma$ tension with SH0ES.

**(c) Late dark-energy transitions.** A third class modifies the expansion history at $z \lesssim 0.1$ through a dark-energy equation of state departing from the CPL form (phantom crossings, sign-switching $\Lambda$, or rapid $w(z)$ transitions). These do *not* touch $r_d$, but generically require two or more tuned parameters and are constrained by SNe Ia and BAO at low $z$; they typically leave a $\sim 1\sigma$ residual.

## 7.2 Comparison table

Table 3 summarises the comparison across seven axes that a referee would use to discriminate the proposals.

Table 3: Quantitative comparison of Hubble-tension resolutions. Values for EDE, SIDR, and late-DE transitions are representative ranges from the literature [Poulin et al., 2019, Smith et al., 2020, Hill et al., 2020, Kreisch et al., 2020, Aloni et al., 2022, Schöneberg et al., 2022]; the SSEE row is the result of this paper. "Free params" counts parameters *beyond* the six of $\Lambda$CDM. The other proposals raise a *predicted* local $H_0$ towards SH0ES; SSEE instead takes the measured local rate as input and predicts the global one, so its residual is that of $H_0^{\rm SH0ES}(1 - f_{\rm screen})$ against the algebraic $3(\varphi + \pi)^2$.

| Axis | $\Lambda$CDM | EDE | SIDR | Late-DE | **SSEE** |
|---|---|---|---|---|---|
| Mechanism epoch | — | $z \sim 3000$ | $z \gtrsim 1000$ | $z < 0.1$ | $z < 0.15$ |
| Modifies $r_d$? | — | Yes ($-$few%) | Yes (via $\Delta N_{\rm eff}$) | No | **No** |
| Free parameters | 0 | 3 | 1–3 | 2+ | **0** |
| Local $H_0$ (km/s/Mpc) | 67.4 | 71–72.5 | 70–71.8 | $\sim 72$ | **input** |
| Residual | $5\sigma$ | $1.6$–$2\sigma$ | $2$–$3\sigma$ | $\sim 1\sigma$ | $\mathbf{0.17\sigma}$ |
| $S_8$ side-effect | — | Worsens | Mild | Variable | **None** |
| CMB peak shift | — | Yes | Yes | No | **No** |

## 7.3 What distinguishes the SSEE mechanism

Three features set the SSEE kinematic screening apart from every entry in Table 3:

(i) **Zero free parameters.** EDE, SIDR, and late-DE transitions each introduce between one and three tunable parameters fitted to data. The SSEE screening fraction $f_{\rm screen} = (\pi - \varphi)/(\varphi + \pi)^2$ is fixed algebraically (Section 3); it is a *prediction*, not a fit. In the language of Schöneberg et al. [2022], the model carries no $\Delta\chi^2$ penalty for added parameters.

(ii) **No early-universe modification.** EDE and SIDR both act before recombination and therefore shift $r_d$ and the CMB peak positions, requiring a delicate re-balancing of $n_s$ and $\omega_{\rm cdm}$ (which is what degrades $S_8$). The SSEE correction is purely a late-time ($z < 0.15$) kinematic effect: $r_d$, $\theta_*$, and the growth factor $\sigma_8$ are untouched (Section 4.3), so the $S_8$ results of Papers 5–6 are preserved.

(iii) **Sub-sigma residual, zero added parameters.** The SSEE output $H_0^{\rm SH0ES}(1 - f_{\rm screen}^{\rm IR})$, compared against the pure number $3(\varphi+\pi)^2$, sits at $0.17\sigma$ from it when the IR cascade is applied to SH0ES — better than every finalist of the $H_0$ Olympics, *and* achieves this with zero added parameters. The UV completion (Paper 10) sharpens it to $0.00\sigma$.

### 7.4 Honest caveats

Two points temper the comparison and are stated explicitly for the referee:

- The screening *fraction* $f_{\rm screen}$ is parameter-free, but the form in which it acts on a measured rate is a postulate (§3.4); the derivations offered for it in earlier versions are withdrawn (§2.2). The competing mechanisms of Table 3 are compared here on their predictions, not on the depth of their derivation.
- As noted in Section 4.2, the cascade run on the TRGB value adopted by Freedman et al. [2025] ($70.39 \pm 1.94$) lands $-1.28\sigma$ from $3(\varphi + \pi)^2$. The spread among local probes is an observational issue independent of any model.

The comparison thus favours SSEE on parsimony and on the absence of CMB and $S_8$ side-effects, while acknowledging that the convergence of local $H_0$ anchors remains an open observational question.

## 8 Limitations and Future Work

We identify two items that lie beyond the scope of this paper.

**Mechanism of the screening (open).** The value of $f_{\rm screen}$ is exact; the form of its action on a measured rate is the postulate (16). Earlier versions listed three arguments for it: a disformal path compression, a separate-universe step, and algebraic consistency. The first two are withdrawn (§2.2); the third fixes the value, not the form. The additive variant differs by $f_{\rm screen}^2 H_0^{\rm glob} \approx 0.31\,{\rm km/s/Mpc}$, below the present SH0ES uncertainty. Two tests of the mechanism were run against present data, with their criteria committed before the run:

(i) *Depth.* A screening confined to a local volume would fade with distance, and the rate read from the Hubble flow would fall toward $H_0^{\rm glob}$ in the deepest supernovae. Fitting the Pantheon+ Hubble flow at $0.023 < z < 0.15$ (277 supernovae, with 77 calibrators) in four depth quartiles, the rate does not fall: the full drop is excluded at $4.70\sigma$ (read the same way, ΛCDM with Planck: $5.12\sigma$), and at most a fraction 0.65 of it is allowed at $3\sigma$. A screening that acts on every local measurement of the rate does not fade with depth and passes.

(ii) *Method.* If the screening acts on every local measurement, the ladder-free methods (masers, lens time delays, sirens) read the local value; if it lives in the Cepheid calibration, they read the global one. With present errors the combined uncertainty of those methods is 2.25 km/s/Mpc, while separating the two readings needs $\approx 1.03$: the test lacks the power to decide, and no verdict is drawn. Criterion 5 of §6 decides it with a single lens.

**UV completion of $M$ (addressed in Paper 10).** The algebraic identity $f_{\rm screen} = s_K/(3\mathcal{M}) = (\pi - \varphi)/(\Omega^2)$ does not depend on $M$ (it cancels in the ratio), so the IR-regime prediction $H_0^{\rm glob,IR} = 68.13\,$km/s/Mpc ($0.17\sigma$ from $3(\varphi+\pi)^2$) is robust. Paper 10 of this series [Almeida, Paper10] derives $M^4 = 45\alpha^2\rho_{\rm crit} = 5\varphi^8\rho_{\rm crit}$ from the same $\alpha$-attractor that governs inflation (Paper 1), yielding $s_K^{\rm full} = 0.41691$ (+3.37% over the IR value) and the UV output the model value $H_0^{\rm glob} = 67.96214\,$km/s/Mpc — not a second $H_0^{\rm glob}$, but the same cascade with $f_{\rm screen}$ complete.

# 9 Conclusions

We have shown that the SSEE equation of state (Paper 7), with the structural constants of the algebra, fixes an algebraically exact screening fraction:

$$f_{\rm screen} = \frac{s_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi+\pi)^2} \approx 0.06725\,, \tag{22}$$

where $s_K = 3\,s_{\rm DE}(1+w_0)$ is the pressure-dilution rate and $\mathcal{M} = \mathcal{A}/2$ is a structural constant. The structural constant $\mathcal{A} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of the two fundamental constants $\varphi$ and $\pi$. The value is algebraic; its multiplicative action on the measured rate is a postulate whose mechanism remains open (§8).

Applied to the measured local rate, the lens gives, in the IR regime,

$$H_0^{\rm glob,IR} = H_0^{\rm SH0ES}\left[1 - \frac{\pi - \varphi}{(\varphi+\pi)^2}\right] \approx 68.13\ \text{km/s/Mpc}\,, \tag{23}$$

which is $0.17\sigma$ from the algebraic $3(\varphi+\pi)^2$ with zero free parameters, the SH0ES datum [Riess et al., 2022] being the input. The earlier MIRA-mapped value 67.037 → 71.87 ($1.12\sigma$) is superseded by the single algebraic target under the 2026 reframe (Paper 1). The correction is a late-time effect ($z < 0.15$) driven by the scalar kinetic energy of the dark energy field; it does not modify the sound horizon $r_d$, the CMB peak structure, or the growth factor $\sigma_8$ established in Papers 2–6.

The result is pre-committed: both values were derived from the algebraic structure of SSEE before this comparison was made. Future SH0ES measurements and upcoming DESI [DESI Collaboration, 2025] data will test this prediction.

**Age of the universe cross-check.** As an independent consistency test, we compute the age of the SSEE universe. The age is a *global* quantity, so the rate that enters is the global one, $H_0^{\rm glob} = 67.962\,$km/s/Mpc—the output of the cascade, not the local $H_0^{\rm SH0ES}$ that is its input—and the density that enters is the one matter density of the model,

$\Omega_m = 0.308881$, with the CPL equation of state [Chevallier & Polarski, 2001, Linder, 2003] $(w_0, w_a) = (-0.840, -0.670)$:

$$t_0 = \frac{1}{H_0}\int_0^\infty \frac{dz}{(1+z)\,E(z)}\,, \qquad E^2(z) = \Omega_m(1+z)^3 + \Omega_{\rm DE}\, f_{\rm DE}(z)\,, \tag{24}$$

where $f_{\rm DE}(z) = (1+z)^{3(1+w_0+w_a)}\exp\!\big(-3w_a z/(1+z)\big)$. Numerical integration gives

$$t_0^{\rm SSEE} = 13.732\ {\rm Gyr} \qquad (\Lambda{\rm CDM\ at\ Planck\ parameters}: 13.801\ {\rm Gyr})\,. \tag{25}$$

The two are indistinguishable: 0.068 Gyr apart, well inside the uncertainty of any current age determination. Both sit comfortably above the oldest globular clusters and halo stars ($t_{\rm GC} \gtrsim 12.5$ Gyr, e.g. Valcin et al. 2021, Bond et al. 2013) and above the Cimatti & Moresco [2023] red-galaxy constraint ($t \approx 13.5$ Gyr at $z \approx 0.55$), so the age does not discriminate between the models in either direction.

**An earlier version of this cross-check, withdrawn.** Previous versions reported $t_0^{\rm SSEE} = 15.52$ Gyr, "1.72 Gyr older than ΛCDM", presented that as more accommodating of old stellar populations, and committed to the criterion "a measurement of $t_0 > 14$ Gyr would favour SSEE". Three inputs were wrong, and they compounded. The integral was fed $\Omega_m = \Omega_{m,\rm dyn} = 0.160050$ as a matter density, but that number is $1 + w_0$, a quantity of the equation of state, and never was a density (Paper 1); it used the *local* $H_0^{\rm SH0ES} = 73.04$ for a global quantity; and it cited the Two-$\Omega_m$ Criterion as its justification, which Paper 1 has since withdrawn. Correcting only the density gives 12.78 Gyr and correcting both gives the value above; the intermediate number is a diagnostic, not a prediction. The stated 15.52 is not even reproducible from the recipe printed alongside it, which returns 15.28. The falsification criterion built on it is withdrawn with it: the model predicts 13.732 Gyr, so a measurement of $t_0 > 14$ Gyr would count *against* SSEE, not for it.

# A Postulate P: the algebraic value SH0ES requires

The body reports the cascade run on the measured local rate $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ km/s/Mpc: with the IR term alone it gives $H_0^{\rm glob,IR} = 68.13$ $(0.17\sigma)$, and with $f_{\rm screen}$ complete (IR+UV, Paper 10) the single model value $H_0^{\rm glob} = 67.96214$ km/s/Mpc, whose residual against $3(\varphi+\pi)^2$ is $+4.2\times10^{-6}$ km/s/Mpc. That residual is *not* a significance: the uncertainty propagated from $H_0^{\rm SH0ES}$ is $\pm0.968$ km/s/Mpc and dominates it by five orders of magnitude. The cascade reduces the $\sim 5\sigma$ Planck–SH0ES discrepancy to below the percent level with zero free parameters.

This appendix supplies the independent self-consistency check that motivates adopting 67.96214 as the algebraic target: it works the screening chain *backwards* from the SH0ES datum, assuming no algebraic value in advance, and recovers exactly the same number.

**Step P1: the inverse chain — what background does SH0ES demand?** The screening relation $H_0^{\rm local} = H_0^{\rm base}/(1 - f_{\rm screen})$ inverts to

$$H_0^{\rm base} \;=\; H_0^{\rm SH0ES}\,(1 - f_{\rm screen}^{\rm UV})\,, \tag{26}$$

which asks, using only the measured datum and the derived screening fraction (no model constant): *what homogeneous background expansion rate would, seen through the SSEE*

*screen* (16)*, reproduce the local SH0ES slope exactly?* With $f_{\rm screen}^{\rm UV} = 0.069522$ from the $X^2/M^4$ correction of Paper 10,

$$H_0^{\rm base} \;=\; 73.04 \times (1 - 0.069522) \;=\; 67.96214\ \text{km/s/Mpc}\,. \tag{27}$$

This number is obtained by deducing *backwards* from the data; $\varphi$ and $\pi$ have not yet been invoked.

**Step P2: the recovered value carries a double algebraic identity.** Only now do we ask whether 67.96214 km/s/Mpc has structure in the closed $(\varphi, \pi)$ dictionary. It does, in two independent ways:

$$H_0^{\rm base} \;=\; \underbrace{3(\pi+\varphi)^2}_{\text{algebraic expression}} \;=\; \underbrace{\textsc{mikael} \times \Omega_{\rm DNAV}}_{\text{SSEE expression}} \;=\; 67.96214\ \text{km/s/Mpc}\,, \tag{28}$$

where $\Omega_{\rm DNAV} = \pi + \varphi = 4.759627$ is the Stability Metric and $\textsc{mikael} = 3\Omega_{\rm DNAV} = 14.278880$ coincides in value with the Maximal Dimensional Invariant $M_v$ (one of the 25 Sovereign paths to $3\Omega$), generated in the lineage from $\textsc{mika} + \pi$ with $\textsc{mika} = K_v + \varphi = 11.1373$. The agreement with the inverse-chain value (27) is $-0.0001\%$.

**Status.** The background that SH0ES requires for exact closure (67.96214) coincides, to five figures, with the pure number $3(\varphi+\pi)^2$ already present in the SSEE dictionary (and independently with $\textsc{mikael} \times \Omega_{\rm DNAV}$). This double identity — recognised a posteriori, not fitted — is the self-consistency evidence that, under the 2026 reframe, justifies adopting 67.96214 as the single algebraic target of the suite rather than the CMB-mapped 67.037 of the earlier register. The dimensional origin of the absolute scale (Postulate D) remains an open problem, as in ΛCDM.

# B The screening postulate and the Hubble-flow integral

We spell out what the postulate (16) implies for the distance integral. This is a consistency statement, not a derivation of the postulate.

**Step 1: the SH0ES measurement is the slope of $d_L(z)$ at $z \to 0$.** The luminosity distance in any FLRW background is

$$d_L(z) \;=\; (1+z)\int_0^z \frac{dz'}{H(z')}\,. \tag{29}$$

For $z \ll 1$, Taylor-expanding $H(z')$ around $z' = 0$ gives

$$d_L(z) \;=\; \frac{z}{H_0^{\rm local}} + \frac{1-q_0}{2}\,\frac{z^2}{H_0^{\rm local}} + \mathcal{O}(z^3)\,, \tag{30}$$

where $H_0^{\rm local} \equiv H(z=0)$ and $q_0$ is the deceleration parameter. The SH0ES programme measures the slope $\mathrm{d}d_L/\mathrm{d}z\big|_{z\to 0} = 1/H_0^{\rm local}$, so:

$$H_0^{\rm local} \;=\; H(z=0)\,. \tag{31}$$

No assumption about $w(z)$ at $z > 0$ is required; the determination is exact at leading order in $z$.

**Step 2: the postulate as a rescaling of the local rate.** The postulate states that the rate a local observer reads near $z = 0$ is $H^{\rm local}(z) = H^{\rm glob}(z)/(1 - f_{\rm screen})$. The comoving distance it implies is

$$\chi^{\rm local}(z) \;=\; \int_0^z \frac{c\,dz'}{H^{\rm local}(z')} \;=\; (1 - f_{\rm screen})\,\chi^{\rm glob}(z)\,. \tag{32}$$

Earlier versions obtained this compression from disformal null geodesics with photon speed $\tilde{c} = c/(1 - f_{\rm screen})$ (Paper 8). The action of Paper 7 has no disformal coupling and $\alpha_T = 0$, so photons travel at $c$: the compression is not a propagation effect and is not derived here. A mechanism that reproduces it is the open problem of §8.

**Step 3: the observed $H_0$ is the global value boosted by $(1 - f_{\rm screen})^{-1}$.** Substituting (32) into $d_L^{\rm obs} = (1 + z)\chi^{\rm local}(z)$:

$$d_L^{\rm obs}(z) \;=\; (1 - f_{\rm screen})\,d_L^{\rm glob}(z) \;=\; (1 - f_{\rm screen})\,\frac{z}{H_0^{\rm glob}} + \mathcal{O}(z^2)\,. \tag{33}$$

A standard FLRW fit to $d_L^{\rm obs} = z/H_0^{\rm local}$ therefore returns

$$\boxed{H_0^{\rm local} \;=\; \frac{H_0^{\rm glob}}{1 - f_{\rm screen}}}\,. \tag{34}$$

This restates the postulate in terms of the observable $d_L$; it is not an independent derivation. The additive approximation $H_0^{\rm glob}(1 + f_{\rm screen})$ follows from the Taylor expansion $(1 - f_{\rm screen})^{-1} \approx 1 + f_{\rm screen}$ and carries a residual error $f_{\rm screen}^2\, H_0^{\rm glob} \approx 0.31\,{\rm km/s/Mpc}$.

**Which end is measured.** Eq. (34) is a relation between two dimensionful rates, and the derivation above says nothing about which of the two is known. That is supplied by observation, not by the model: the distance ladder measures $H_0^{\rm local}$, so the relation is used in the form $H_0^{\rm glob} = H_0^{\rm local}(1 - f_{\rm screen})$ throughout this paper. The dimensionless $3(\varphi + \pi)^2$ never enters the cascade; it is what the output is compared against.

**Step 4: $f_{\rm screen}$ is exactly constant.** The compression (32) uses a $z$-independent $f_{\rm screen}$. With the corrected identification this needs no approximation: $s_K = -3w_0(1 + w_0)$ is built from $w_0$ alone, and $w_0$ is a constant of the model, so $s_K$ is a number and not a function of redshift. Since $\mathcal{M}$ is a structural constant, $f_{\rm screen} = s_K/(3\mathcal{M})$ is constant *exactly*, at all $z$.

Earlier versions of this paragraph read $\alpha_K(z) = 3\Omega_{\rm DE}(z)(1 + w(z))$ and found a 56% variation over $z \in [0, 0.15]$, which had then to be argued away as formally of order $w_a z_{\rm max}$. That variation belonged to the kineticity, which does evolve; it never belonged to $s_K$. The correction removes the caveat rather than weakening it.

A structural check makes the same point from the other side. Replacing $s_{\rm DE} = T_r/M_v$ by the density fraction $1 - \omega_m/h^2$ in Eq. (11) makes $f_{\rm screen}$ depend on the expansion anchor: sweeping that anchor from 60 to 100 $\rm km\,s^{-1}Mpc^{-1}$ moves the result. With $s_{\rm DE}$ it is invariant to the last digit, because no $H$ enters. A screening fraction that moved with the anchor used to derive $H_0$ would be circular; this one cannot be.

The mean value of $f_{\rm screen}(z)$ over this interval is approximately

$$\bar{f}_{\rm screen} \;\approx\; f_{\rm screen}(0)\left[1 + \frac{3w_a\, z_{\rm max}}{4}\right] \;\approx\; 0.06725 \times \big(1 - 0.025\big) \;\approx\; 0.0656\,, \tag{35}$$

a 2.5% correction to $f_{\rm screen}$ that shifts $H_0^{\rm local}$ by $< 0.05\,\mathrm{km/s/Mpc}$—well below the $1.04\,\mathrm{km/s/Mpc}$ SH0ES uncertainty. The $z = 0$ value $f_{\rm screen} = 0.06725$ is therefore the correct representative for the SH0ES measurement, and equation (34) is accurate to 0.1% within the SH0ES calibration volume.

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